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| **Fiscal Year:** | FY 2019 | **Task Last Updated:** | FY 01/11/2021 |
| **PI Name:** | Ploutz-Snyder, Lori L. Ph.D. | | |
| **Project Title:** | Integrated Resistance and Aerobic Training Study (Sprint) | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Aerobic**:Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity<br>(2) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone<br>(3) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance<br>(4) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | lorips@umich.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (734) 764-5210 |
| **Organization Name:** | University of Michigan | | |
| **PI Address 1:** | OBL 4170, 1402 Washington Hts. | | |
| **PI Address 2:** | School of Kinesiology | | |
| **PI Web Page:** | | | |
| **City:** | Ann Arbor | **State:** | MI |
| **Zip Code:** | 48109-2013 | **Congressional District:** | 12 |
| **Comments:** | Previously at Universities Space Research Association/NASA Johnson Space Center until July 2016. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 10/01/2016 | **End Date:** | 01/31/2019 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
| **Contact Email:** | Peter.norsk@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 1/31/2019 per NSSC information and J. McFather/JSC (Ed., 7/8/19) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Ploutz-Snyder, Robert Ph.D. ( University of Michigan ) | | |
| **Grant/Contract No.:** | NNX16AO74G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | ED. NOTE (October 2016): Continuation of "Integrated Resistance and Aerobic Training Study (Sprint)," which was an internal project at Universities Space Research Association/NASA Johnson Space Center, while Principal Investigator was affiliated there.<br>Current exercise countermeasures are insufficient to prevent muscle atrophy, cardiovascular deconditioning, and bone loss associated with long-duration spaceflight. A known limitation has been the inability of the International Space Station (ISS) exercise hardware to provide sufficient loads to the human body that are required for maintaining physiological function. New flight exercise hardware, including the advanced resistance exercise device (ARED) and 2nd generation treadmill (T2), are designed to provide astronauts with the ability to exercise at higher intensity. This opens an array of new possibilities for exercise programming in long-duration spaceflight. While the ability to do resistance exercises at heavier loads, the ability to run faster and potentially with more body loading are obvious improvements in exercise capabilities, the details of how this new equipment should be used are not so obvious. Towards this end, two workshops were held, the NASA Muscle Workshop in June 2008 and the NASA International Space Station Exercise Prescription Workshop in October 2008. Intramural and extramural experts concluded that using higher intensity resistance exercises and interval aerobic exercise would help to maintain physiological function while simultaneously decreasing total exercise time and volume. The panel then recommended that this evidence-based approach to the development of a novel exercise prescription for use on ISS should immediately be proposed. Therefore the purpose of this research is to evaluate the efficacy of a new integrated resistance and aerobic training (iRAT) program designed to minimize loss of muscle, bone, and cardiovascular function during ISS missions.<br><br>Highlights of the iRAT include an increase in the intensity and a reduction in the volume of resistance exercises, inclusion of very short, but high intensity interval-type aerobic exercises, and starting the exercise countermeasures as early as possible in the flight, preferably in the first week. Pre-, in-, and post-flight testing, and data sharing with selected on-going medical assessment tests, will be used to assess the effectiveness of this candidate prescription. Tests include detailed measurements of lower body muscle structure and function (MRI, ultrasound, muscle performance testing, shared muscle medical tests), maximal aerobic capacity, ventilatory threshold, left ventricular mass and cardiac contractility, and bone mineral density. Twenty (20) long duration crew members will be recruited as active subjects in iRAT; and 20 will be asked to serve as controls that perform the standard care exercise prescription and the pre-post flight testing.<br><br>This proposal is sponsored by the Exercise Physiology and Countermeasures Project and directly addresses the Human Research Program (HRP) Integrated Research Plan (IRP) Risks of Impaired Performance Due To Reduced Muscle Mass, Strength and Endurance and Risk of Reduced Physical Performance Capabilities Due To Reduced Aerobic Capacity, and Risk of Accelerated Osteoporosis. This proposal addresses Gap M7: Can the current in-flight performance be maintained with reduced exercise volume? The proposal addresses IRP Gap M8 (What is the minimum exercise regimen needed to maintain fitness levels for operationally relevant tasks?) and Gap M9 (What is the minimum set of equipment needed to maintain those (M8) fitness levels?). |
| **Rationale for HRP Directed Research:** | This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. |
| **Research Impact/Earth Benefits:** | Development of more efficient exercise programs could be used on Earth.<br>A more thorough understanding of how cardiovascular, skeletal muscle, and bone health is affected by exercise. |
| **Task Progress:** | Ed. note: Reporting not received and considered "done" per Element management. See Bibliography for publications from the project. |
| **Bibliography Type:** | Description: (Last Updated: 11/21/2025) |
| **Articles in Peer-reviewed Journals** | Ryder JW, Fullmer P, Buxton RE, Crowell JB, Goetchius E, Bekdash O, DeWitt JK, Hwang EY, Feiveson A, English KL, Ploutz-Snyder LL. "A novel approach for establishing fitness standards for occupational task performance." Eur J Appl Physiol. 2019 Jul;119(7):1633-48. https://doi.org/10.1007/s00421-019-04152-3 ; PMID: 31069517. , Jul-2019 |
| **Articles in Peer-reviewed Journals** | English KL, Downs M, Goetchius E, Buxton R, Ryder JW, Ploutz-Snyder R, Guilliams ME, Scott JM, Ploutz-Snyder LL. "High intensity training during spaceflight: Results from the NASA Sprint Study." npj Microgravity. 2020 Aug 18;6(1):21. https://doi.org/10.1038/s41526-020-00111-x ; PMID: 33574275 , Aug-2020 |
| **Articles in Peer-reviewed Journals** | Keller N, Whittle RS, McHenry N, Johnston A, Duncan C, Ploutz-Snyder L, De La Torre GG, Sheffield-Moore M, Chamitoff G, Diaz-Artiles A. "Virtual Reality "exergames": A promising countermeasure to improve motivation and restorative effects during long duration spaceflight missions." Front Physiol. 2022 Oct 11;13:932425. https://doi.org/10.3389/fphys.2022.932425 ; PMID: 36304582; PMCID: PMC9593063 , Oct-2022 |
| **Articles in Peer-reviewed Journals** | Trappe TA, Minchev K, Perkins RK, Lavin KM, Jemiolo B, Ratchford SM, Claiborne A, Lee GA, Finch WH, Ryder JW, Ploutz-Snyder L, Trappe SW. "NASA SPRINT exercise program efficacy for vastus lateralis and soleus skeletal muscle health during 70 days of simulated microgravity." J Appl Physiol (1985). 2024 May 1;136(5):1015-39. https://doi.org/10.1152/japplphysiol.00489.2023 ; PMID: 38328821; PMCID: PMC11365553 , May-2024 |
| **Articles in Peer-reviewed Journals** | Skiles CM, Boyd G, Gouw A, Robbins E, Minchev K, Ryder J, Ploutz-Snyder L, Trappe TA, Trappe S. "Myonuclear and satellite cell content of the vastus lateralis and soleus with 70 days of simulated microgravity and the NASA SPRINT exercise program." J Appl Physiol (1985). 2025 Jan 1;138(1):195-202. https://doi.org/10.1152/japplphysiol.00468.202 ; PMID: 39656504; PMCID: PMC12188944 , Jan-2025 |